Homework 4: 3D Transformations

CS 440 Computer Graphics
Habib University
Spring 2026

Due: 11:59pm on Friday, 13 March

For the problems in this assignment, you may use the `webgl-utils.js` and `initShaders.js` files provided in the recitations. It is also strongly recommended that you use your `common.js` file that you created as part of your first assignment. You may create any new helper functions as you wish in `common.js`.

For questions in this assignment, you will need to load mesh files that are stored locally. Browsers usually do not allow JS to load a local file for security purposes, and attempting to do so may result in a CORS (cross-origin) error. The simplest way to work around this is to run a local server using Python, as described below. If you choose to use any other method, please clearly specify it in a `README` file.

- Open your terminal and go to your working directory. Run the command `python -m http.server`.
- After running this command, you should see a message such as `Serving HTTP on :: port 8000....` The port number may vary depending on your system, but you should see a similar message indicating that the server is running.
- Open your browser and navigate to `http://localhost:8000/<filename>`. You should now be able to access your local files.

1. Meshes

Browse the The Stanford 3D Scanning Repository and download any (or all!) of the 3D models.
Task: Write a program to load a mesh from file and transform it to WebGL's default clip coordinates such that the aspect ratio of the model is preserved and the model is centered at the origin. The program should provide the user the means to interact in the following ways with the model.

(a) Rotate the model slightly about each of the 3 axes in the clockwise or counterclockwise direction.
(b) Shift the model slightly in the positive and negative x, y, or z directions.
(c) Shrink or grow the model slightly along any of the x, y, or z directions.
(d) Reflect the model about any of the axes.
(e) Apply a positive or negative shear to the model along any of the axes
(f) Reset any transformations that have been applied to the model.

Files: `mesh.html, mesh.js`

2. Bonus: Fun with Platonic Solids

A platonic solid is a 3D polyhedron where every face is an identical regular polygon (equilateral triangle, square, or pentagon) and the same number of faces meet at each vertex. The dual of a Platonic solid is obtained by interchanging its vertices and faces. For example, the dual of a tetrahedron is another tetrahedron, while the dual of a cube is an octahedron. There are five such polyhedra, as summarized in the table below.

Platonic Solid	**Dual**	**Vertices**	**Faces**
Tetrahedron	Tetrahedron	4	4
Hexahedron (Cube)	Octahedron	8	6
Octahedron	Hexahedron (Cube)	6	8
Dodecahedron	Icosahedron	20	12
Icosahedron	Dodecahedron	12	20

Task: You are provided .PLY files[1] (indexed mesh representation) for the five platonic solids. Load the files and display them in your WebGL scene. You should only show one platonic solid at once. Your program should also support the following interactions:

- Pressing 1-5 should switch between different platonic solids (according to the order listed in the table above).
- Press D to switch between a platonic solid and its dual.
- Press W to toggle the display of mesh edges (wireframe), rendered in black.

Note that the duals must be computed by your program, you cannot load them from a file. This will require you to calculate centroids for each face (average of all vertex positions of a face), which is fairly easy. You will also need to loop through all the adjacent faces around a given vertex in order to to create a new polygon that replaces each vertex. For this, you can either directly sort face centroids, or define a mesh data structure such as half-edge.

Files: `solids.html, solids.js`

Submission Instructions

You will turn in your code via Canvas. Please submit a single zip file containing all the required files. The zip file must be named using your HU ID. For example, if my ID is aa04049, I would name my zip file as "04049.zip". Please **do not** include your name or initials anywhere in your submission. Also, do not create a .rar or .tar file – please create a .zip file.
Make sure to follow the specified file naming convention. Also make sure that you create separate files for each problem as specified. If you do not follow the submission instructions it is possible that you will get a small point deduction.

[1] Obtained and modified from FSU PLY repository.